Daniel Garcia Bruña

+34 618153131 | garciabrunadaniel@gmail.com | linkedin.com/in/danielgarciabruna

Education

TU Delft — Delft, NL
Pre-Master in Applied Mathematics — *Sep. 2026 – Jun. 2027*
Preparation for M.Sc. Applied Mathematics

- *Coursework in Real Analysis, Partial Differential Equations, Measure Theory, and Probability Theory.*

Erasmus University Rotterdam — Rotterdam, NL
B.Sc. in Econometrics and Operations Research — Upper 2:1 — *Sep. 2023 – Sep. 2026*
Minor in Computer Science

- *Coursework in optimization, stochastic processes, statistics, algorithms, data mining, and advanced programming.*

Rotterdam School of Management — Rotterdam, NL
B.Sc. in International Business Administration — First Class — *Sep. 2022 – Sep. 2025*

Experience

Quantitative Analyst Intern | *ASML* — Feb. 2026 – Jun. 2026

- Develop stochastic decision models to estimate the expected ROI of workforce training programs under uncertainty in demand, productivity, and learning curves.
- Design cost–benefit frameworks incorporating opportunity cost, training duration, and operational constraints to compare alternative training strategies.
- Implement a modular Python decision engine using structured data models to evaluate training scenarios, perform sensitivity analysis, and generate management-ready cost comparisons.

Projects

High-Dimensional Forecasting | *Python, NumPy, pandas, scikit-learn* — 2025

- Built a scalable end-to-end forecasting pipeline for high-dimensional retail sales across hundreds of SKUs and multiple forecast horizons, supporting automated data ingestion, preprocessing, and model deployment.
- Engineered features using time series decomposition, structured lag hierarchies, and calendar effects, and trained regularized linear and elastic net models to control overfitting in sparse, correlated feature spaces.
- Evaluated model performance using rolling-origin cross-validation and error decomposition, emphasizing forecast stability, coefficient sparsity, and robust out-of-sample generalization.

Hybrid LSTM Volatility Forecasting | *Python, PyTorch, time series* — 2025

- Implemented hybrid LSTM architectures in PyTorch combining GARCH-type volatility components with HAR-RV and VIX-based features for multi-horizon volatility forecasting.
- Benchmarked hybrid neural forecasts against standalone GARCH-family and HAR models across assets and market regimes using rolling-window evaluation and risk-relevant error metrics.
- Tested statistical significance of performance differences and feature contributions using hypothesis testing and sensitivity analysis in low-signal, highly correlated feature spaces.

Additional Activities

Optiver Hackathon — Nov. 2025

- Achieved a top-three finish in a university hackathon by developing a market-making strategy for a single product traded across two correlated markets.

IMC Prosperity Trading Challenge — 2025

- Participated in an algorithmic trading competition by IMC Trading in a 3-week simulated trading environment.
- Developed and tested strategies in market making, option mispricing exploitation, arbitrage, and mean reversion.
- Ranked 4th in NL and 103rd globally out of 5,000 participants before withdrawing due to overlapping exams.

Skills

Programming Languages: Python, R, Java, SQL
Languages: Spanish (Native), English (Bilingual), German (Intermediate)